

Referee Report

Is Spurious Regression a Problem in Stationary Time Series? A Theoretical Framework and Monte Carlo Evidence

Reviewer

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Summary of the submission

The paper asks whether the Granger–Newbold spurious regression problem survives under stationarity. Its answer is a clean and correct dichotomy: stationarity plus a zero population projection slope buys *consistency* of $\hat{\beta}$, but it buys nothing at all for the *conventional standard error*. The limiting conventional t -ratio is $N(0, \lambda)$ with

$$\lambda = \frac{\text{LRV}\{(x_t - \mu_x)u_t\}}{\text{Var}(x_t)\text{Var}(u_t)},$$

so the test over-rejects, under-rejects, or is correctly sized according to whether $\lambda \geq 1$. For independent AR(1) series $\lambda = (1 + \phi_x \phi_y)/(1 - \phi_x \phi_y)$, which depends only on the *product* of the two persistence parameters — hence the opposite-sign case under-rejects severely. Monte Carlo evidence covers eight DGPs plus three extensions (common stochastic volatility, local-to-unity triangular arrays, and distributed-lag Granger tests under omitted dynamics).

Assessment. The core theory is right, the derivations I checked are correct, and the Monte Carlo design is unusually careful in several respects — stationary initialisation from the invariant distribution, nested prefixes, an explicit Monte Carlo standard error, a single-source computational script with content-hashed caching, and a bandwidth sequence whose rate conditions are stated rather than assumed. Every numerical claim in the Results section reconciles with the CSV output files, which is more than can be said for most submissions.

However, there are **two substantive errors** that must be fixed. One is a theoretical claim in §3.2 that is directly contradicted by the paper’s own §3.5. The other is an empirical claim in §5.2 that is directly contradicted by the paper’s own Table 2. There are also two compilation bugs that silently drop references from the bibliography, a miscited reference, and a set of design gaps that keep this a competent replication of known results rather than a contribution. My recommendation is **major revision**.

I have verified the analysis against the source .qmd, the R script, the seven CSV output files, and the compiled PDF.

Major points

M1. The stated sufficient condition for $\lambda = 1$ is wrong, and the paper refutes it itself

§3.2 asserts:

“Serial uncorrelatedness of w_t is sufficient for $\lambda = 1$, but it is not necessary.”

This is false. Decompose the long-run variance into its lag-zero and serial parts:

$$\lambda = \underbrace{\frac{\mathbb{E}[(x_t - \mu_x)^2 u_t^2]}{\sigma_x^2 \sigma_u^2}}_{\kappa} + \frac{2}{\sigma_x^2 \sigma_u^2} \sum_{k \geq 1} \text{Cov}(w_t, w_{t+k}).$$

If w_t is serially uncorrelated then $\lambda = \kappa$, and $\kappa = 1$ **only** if $\mathbb{E}[(x_t - \mu_x)^2 u_t^2] = \mathbb{E}[(x_t - \mu_x)^2] \mathbb{E}[u_t^2]$ — that is, only under conditional homoskedasticity of the score. Serial uncorrelatedness alone is not sufficient.

The paper’s own §3.5 is the counterexample. There, $x_t = s_t \varepsilon_t$ and $y_t = s_t \eta_t$ give a score $w_t = s_t^2 \varepsilon_t \eta_t$ that is *exactly* serially uncorrelated at every nonzero lag, yet $\lambda = \kappa = e^{\nu_h} > 1$. §5.3 states this explicitly: “a conventional size distortion can arise from contemporaneous second-moment dependence even when the score has no serial autocovariance.” The two sentences cannot both stand.

The same error recurs in the Discussion, fourth paragraph: “a zero net sum of nonzero-lag autocovariances produces correct asymptotic size.”

Required fix. Replace both sentences with the correct condition, $\lambda = 1 \iff \Omega = \sigma_x^2 \sigma_u^2$, and state that this requires *both* a zero net serial sum *and* $\kappa = 1$. Serial uncorrelatedness plus score conditional homoskedasticity (e.g. u_t independent of x_t) is the correct sufficient condition.

Opportunity. The κ decomposition above is not merely a patch — it is the paper’s best available upgrade. It unifies three results the paper currently presents as separate episodes: the AR(1) formula ($\kappa = 1$, all distortion serial), the linear-process formula ($\kappa = 1$), and the common-volatility example ($\kappa = e^{\nu_h}$, zero serial term). Promoting it to a displayed lemma early in §3.2 would give the paper a single organising identity and would make §3.5 a corollary rather than an appendix-like aside.

M2. §5.2 overstates HAC performance, contradicting Table 2

§5.2 claims:

“HAC inference moves rejection rates toward 5% in **every** serially correlated case, including both over-rejection and under-rejection.”

Table 2 of the same paper reports, for $(\phi_x, \phi_y) = (0.5, 0.5)$ at $T = 50$: conventional 0.112, HAC 0.136. The conventional test is *closer* to nominal size ($|0.112 - 0.05| = 0.062$ vs. $|0.136 - 0.05| = 0.086$). HAC makes it worse. The word “every” is false.

Required fix. Qualify to something like “for $T \geq 200$ in every serially correlated case; at $T = 50$ the HAC correction is dominated by long-run variance estimation noise and can be worse than the uncorrected test.”

M3. The Granger HAC result plateaus rather than converging, and the text obscures it

§5.5 says HAC “reduces the rejection rate to 7.2% at $T = 2000$, but its small-sample performance is imperfect.” The full sequence for the misspecified model is 0.190, 0.095, 0.080, 0.072, 0.072 at $T = 100, 250, 500, 1000, 2000$. It is flat at 7.2% across a doubling of the sample. The Monte Carlo standard error at $N = 1000$ and $p \approx 0.072$ is 0.8 percentage points, so the stall is not attributable to noise over that one comparison, though it is not decisively established either. “Small-sample performance is imperfect” reads as a claim about $T = 100$; the honest statement is that convergence to nominal size is *not demonstrated within the reported range*.

Relatedly, the framing in §3.8 invites a misreading. The text warns that HAC “does not correct a biased coefficient caused by genuine omitted-variable correlation.” True in general — but in *this* experiment x and y are generated independently, so the omitted y_{t-1} is orthogonal to x_{t-1}, x_{t-2} and the projection coefficients remain exactly zero. There is **no bias here at all**; the entire 30–34% rejection rate is a variance-estimation failure, which is precisely why HAC does most of the repair. Say so explicitly, or a reader will take the experiment as evidence for a claim it does not test.

M4. Missing literature, including the paper’s closest antecedent

The paper’s central result is not new, and the bibliography does not acknowledge the work that established it. This is the most serious scholarly gap. At minimum:

- **Granger, Hyung & Jeon (2001)**, “Spurious regressions with stationary series,” *Applied Economics* 33(7). This is the direct antecedent — same question, same AR(1) over-rejection finding. Its absence is conspicuous and a reader familiar with the literature will assume it was overlooked rather than superseded.
- **Ventosa-Santaularia (2009)**, “Spurious Regression,” *J. Probability and Statistics* — the standard survey; would let the paper position itself in one paragraph.
- **White (1980)** for heteroskedasticity-consistent covariance estimation (see M5).
- **Kiefer & Vogelsang (2002, 2005)** on fixed- b asymptotics, and **Lazarus, Lewis, Stock & Watson (2018)**, *JBES*, “HAR inference: recommendations for practice.” These bear directly on §5.2: the small- T HAC over-rejection the paper documents is the canonical motivation for fixed- b critical values, and LLSW give explicit bandwidth guidance that differs sharply from the rule used here.
- **Andrews (1991)** and **Andrews & Monahan (1992)** for data-dependent bandwidth and prewhitening — the natural remedies for the (0.95, 0.95) case that the paper flags but does not treat.
- **Cavanagh, Elliott & Stock (1995)**, **Elliott (1998)**, and **Stambaugh (1999)** for §3.6. Non-uniformity near the unit root and its consequences for predictive regression is a large literature; citing only Phillips (1987) understates it considerably.

hamilton1994time sits in references.bib uncited. Harmless, but tidy it up.

M5. Engle (1982) is miscited for heteroskedasticity-consistent standard errors

The Introduction attributes HC and HAC standard errors to [Engle1982autoregressive; NeweyWest1987]. Engle (1982) introduced ARCH — a *model* of conditional heteroskedasticity, not a covariance estimator. The correct citation is **White (1980)**. As it stands the Introduction misattributes a foundational result, and the rendered bibliography lists Engle for a contribution that is not his.

M6. All experiments are size experiments; there is no power evidence

Every reported quantity is a rejection frequency under a true null. The paper’s central practical recommendation — use HAC — therefore rests on only half the relevant evidence. HAC buys size at a known cost in power, and that cost is largest exactly where the paper’s size gains are largest (high ϕ , large bandwidth). Without size-adjusted power curves against a local alternative $\beta_0 = b/\sqrt{T}$, a reader cannot tell whether the recommendation is a good trade at $\phi = 0.95$ — where the HAC test is *still* 10.7% oversized at $T = 5000$ and has presumably paid substantial power for the privilege.

This is the single most valuable addition available. Adding local power to the existing framework costs little: the simulation machinery already exists, and only the DGP needs a drift term.

M7. The Monte Carlo never tests the paper’s most interesting theoretical claim

§3.4 derives $\lambda = \sum_k \rho_x(k) \rho_y(k)$ for general independent linear processes and correctly observes that this “covers short-memory ARMA processes, unequal lag structures, seasonal dependence, and negative autocorrelation.” None of these appear in the simulation. Every stationary DGP is an AR(1) with unit innovation variance.

The gap matters most for the “sufficient but not necessary” claim in §3.2, which is asserted and never demonstrated. A single extra DGP would fix this: choose ARMA or MA processes whose autocorrelation functions satisfy $\sum_k \rho_x(k) \rho_y(k) = 1$ with both series visibly serially correlated. Showing a *correctly sized* conventional test on visibly autocorrelated data would be the most memorable exhibit in the paper, and it directly serves the paper’s thesis that serial correlation per se is not the diagnostic.

An MA(1) pair with opposite-signed coefficients would also usefully separate “persistence” from “positive autocorrelation” in a way the AR(1) grid cannot.

M8. HAC implementation is not consistent across experiments

Two different HAC estimators are used, and the paper does not say so:

- §5.1–5.4 use a hand-coded Bartlett estimator (`simple_slope_tests`, and the inline loop in `simulate`) with divisor T , **no** degrees-of-freedom adjustment.
- §5.5 uses `sandwich::NeweyWest(..., adjust = TRUE)`, which *does* apply a $n/(n-k)$ finite-sample correction.

At the sample sizes considered the difference is second-order, but it is a comparability defect across the paper’s own tables and it undermines the “single computational source” claim in §4. Standardise on one, and state the choice.

Two related points. First, the hand-coded HAC t -statistic is compared against t_{n-2} critical values while Proposition 3.8 gives an $N(0, 1)$ limit; the mismatch is immaterial asymptotically but should be stated. Second, the bandwidth $L_T = \lfloor 2T^{1/3} \rfloor$ is far more aggressive than the Newey–West rule of thumb $4(T/100)^{2/9}$ — 34 versus 9 at $T = 5000$. The script’s comment defends this, but the defence belongs in the paper, and it should be accompanied by a sensitivity check across at least two bandwidth rules and

one alternative kernel (QS). At present a reader cannot distinguish “HAC works” from “this particular bandwidth works.”

M9. Local-to-unity initialisation is consequential and undocumented

`draw_ar1(n, phi)` initialises from the invariant distribution $N\{0, 1/(1 - \phi^2)\}$. Under $\phi_T = 1 - 5/T$ this means a starting value with standard deviation $\approx \sqrt{T/10}$ — at $T = 5000$, roughly 22. This is the *stationary-start* local-to-unity design, whose limit is an Ornstein–Uhlenbeck process begun in its stationary distribution. That is a defensible choice, but it is a choice: the zero-start design gives materially different finite-sample rejection frequencies. §4 should state it, and ideally report both.

While there, the paper can sharpen its own claim. §5.4 says conventional rejection rises “rather than approaching a fixed stationary plateau.” The stronger and correct statement is that under local-to-unity asymptotics with c fixed, $\hat{\beta}$ has a nondegenerate limit and the t -ratio diverges at rate $\sqrt{T}$, so $\Pr(\text{reject}) \rightarrow 1$ — the *same* qualitative outcome as the unit-root benchmark. The observed 94.1% at $T = 5000$ against the $I(1)$ benchmark’s 97.0% makes the point vividly, and the paper has the number in hand but never reports it.

M10. Reported evidence is thinner than the evidence generated

Three concrete omissions:

1. **The $I(1)$ benchmark appears only as a line in Figure 1.** It is excluded from Table 1 by the merge with `stationary_spec`. Its rejection frequencies (69.1%, 88.4%, 97.0% at $T = 50, 500, 5000$) are the paper’s reference point for “what real spurious regression looks like” and belong in a table.
2. **No Monte Carlo standard errors in any table.** §4 helpfully computes the maximum MCSE (1.12pp) but then reports every entry to three decimals, implying a precision the design does not deliver. Either report $\pm$ MCSE per cell or round to two decimals. The (0.8, −0.8) row reads “0.000” for an asymptotic value of 2.9×10^{-5} — report it in scientific notation or as “< 0.0001”, since “0.000” invites the reader to think it is exactly zero.
3. **No direct check of the distributional claim.** Proposition 3.5 states $t_{OLS} \Rightarrow N(0, \lambda)$, but only tail rejection rates are reported. The sample *variance* of the t -statistics across replications should converge to λ — a one-line addition to the existing loop that would test the proposition far more sharply than a 5% tail probability, and would distinguish “right variance” from “right tail by coincidence.” A QQ plot for one or two DGPs would be a natural figure.

I would also note that the paper’s Introduction invokes Granger and Newbold’s R^2 and Durbin–Watson diagnostics but never simulates them. Reporting mean R^2 across the stationary DGPs would show it collapsing toward zero while rejection rates stay at 35–66% — a compact demonstration that the stationary pathology is *not* classical spurious regression, which is exactly the paper’s thesis.

Compilation and presentation defects

These are mechanical, but two of them corrupt the rendered document.

P1. Citations inside LaTeX environments are not processed — two references vanish

Assumption 3.2 in the .qmd contains `[@davidson1994stochastic; @davidson2000econometric]` inside a raw `\begin{assumption}` block. Pandoc does not process Markdown inside raw LaTeX, so **the compiled PDF prints the literal string** `[@davidson1994stochastic; @davidson2000econometric]`

on page 3, and **both Davidson references are absent from the bibliography** (verify: the reference list runs Engle, Granger, Granger–Newbold, Newey–West, Phillips, Phillips — no Davidson).

Fix by using `\cite{davidson1994stochastic,davidson2000econometric}` inside the environment, or by moving the citation to a Markdown paragraph immediately following it.

P2. Every assumption cross-reference in the prose is wrong

`\newtheorem{theorem}{Theorem}[section]` numbers within sections, so the assumptions render as **3.1, 3.2, 3.3**. The prose refers to them as “Assumption 1”, “Assumptions 1–2”, “Assumption 3” in at least seven places (§3.1 after Assumption 3.3; Propositions 3.4, 3.5, 3.8; Theorem 3.9, twice). Every one of these is a dangling reference in the compiled document.

Fix by dropping `[section]` from the `\newtheorem` declaration, or by adding `\labels` and using `\ref` throughout. The latter is safer.

P3. Figures overflow the text block and waste most of their width

All five figures are 7 in or 6.5 in wide at out-width: 100%, against an A4 text block of $8.27 - 1.2 - 1.0 = 6.07$ in given the stated margins. In the compiled PDF, Figure 2 visibly extends past both margins.

More consequentially, Figures 2–5 plot T on a **linear** axis with breaks at 50, 200, 500, 1000, 2000, 5000. Everything interesting happens below $T = 1000$, which is compressed into the leftmost fifth of the panel, while the $2000 \rightarrow 5000$ interval — where nothing changes — consumes 60% of the width. `scale_x_log10()` (or `scale_x_continuous(trans="log10")`) would fix this immediately and make the convergence behaviour legible. It would also remove the need for the 30-degree rotated tick labels, which currently collide.

Figure 2’s “Standard error” legend is also clipped at the right edge: the “HAC” key label is cut off in the compiled output.

P4. Hard-coded axis limits can silently delete data

`scale_y_continuous(limits = c(0, 0.25))` and friends **drop** out-of-range observations rather than clipping the view, and `warning: false` in the YAML suppresses the `geom_line` removal warning that would otherwise flag it. The current values happen to fit, but a rerun with a different seed, a higher ϕ , or a larger c in the local-to-unity design would silently lose points. Use `coord_cartesian(ylim = ...)` throughout.

More generally, `execute: warning: false` combined with `message: false` means the document will compile quietly through font-fallback failures and near-singular `solve()` calls in `joint_wald_reject`. Set `warning: true` while developing.

P5. Minor presentation

- Table 1’s header reads `lambda` in plain text; it should be λ .
- Table 3’s `RobustMethod` should be “Robust method”, and the table mixes HC0 and HAC in a single “Robust” column without reporting the HAC value for the common-volatility row.
- “References” is numbered as §7 and the Computational Appendix as §8. Conventionally the reference list is unnumbered and appendices are lettered.
- The document has no abstract, no keywords, and no JEL codes. For a paper of this length and register, an abstract is expected.
- `mainfont: "Times New Roman"` is proprietary and absent from most TeX installations; TeX Gyre Termes is the portable equivalent. Correspondingly, `theme_classic(base_family`

- = "Times") in the R script will fall back silently on many systems. Math is set in Latin Modern against Times body text, which is visible on every display equation; set a matching `mathfont`.
- Cross-referencing style is mixed: Quarto `@fig-@tbl-` references alongside raw LaTeX `\ref{}` for propositions. The `\ref` calls will not resolve if the document is ever rendered to HTML.
- The QED square after the Proposition 3.7 proof floats onto its own line, well below the display. Add `\qedhere` or restructure the final line.

Reproducibility

The single-source script with `cache.extra = md5sum(...)` is good practice and I want to credit it. The remaining issues are straightforward:

- **Tmax is captured from the global environment** inside `simulate_common_volatility()`, which does not take it as an argument (line 194). The function is not self-contained and will fail or silently misbehave if called from a different scope.
- **set.seed(2026) is called once**, before the main loop only. The three extension experiments inherit whatever RNG state the main loop leaves behind, so changing `Nsim` or the DGP list silently changes every extension result. Seed each experiment independently.
- **sandwich is used via :: but never declared**. Neither is the R version, package versions, nor `sessionInfo()`. Add a dependency block or an `renv` lockfile; a paper whose central claim is numerical should pin its numerical environment.
- **T <- grid_shadows base::T** inside `simulate()`. It works, but shadowing `T` in R is a well-known footgun and reviewers will flag it.
- `library(knitr)` is loaded but unused in the script.
- `joint_wald_reject` calls `solve(V, b)` with no singularity guard.
- Housekeeping: `.DS_Store`, `Rplots.pdf` (a side effect of the `print()` calls at the foot of the script), and an empty `comments.json` are all committed.

Verified correct

For the record, I checked the following and found no error:

- **Proposition 3.4** (consistency). Correct; ergodicity of the second moments is sufficient.
- **Proposition 3.5**. The algebra is right: $T \text{se}^2 \rightarrow_p \sigma_u^2 / \sigma_x^2$ combines with $\sqrt{T} \hat{\beta} \Rightarrow N(0, \Omega / \sigma_x^4)$ to give $t \Rightarrow N(0, \lambda)$ with $\lambda = \Omega / (\sigma_x^2 \sigma_u^2)$.
- **Proposition 3.7**. $\text{Cov}(w_t, w_{t+k}) = \gamma_x(k) \gamma_y(k) = (\phi_x \phi_y)^{|k|} \sigma_x^2 \sigma_y^2$, summing to $\lambda = (1 + \phi_x \phi_y) / (1 - \phi_x \phi_y)$. Two notes: the proof writes the exponent as k where it should be $|k|$; and the Gaussianity assumption is unnecessary — mutual independence of the two innovation *processes* plus finite fourth moments suffices, and dropping Gaussianity would strengthen the proposition at no cost.
- **§3.4 linear-process formula**. $\lambda = \sum_k \rho_x(k) \rho_y(k)$ is correct.
- **§3.5 common-volatility calculation**. With $h_t \sim N(0, v_h)$: $\mathbb{E}[s^2] = e^{v_h/2}$, $\mathbb{E}[s^4] = e^{2v_h}$, so $\lambda = e^{v_h}$. Correct. The score is exactly serially uncorrelated, as claimed.
- **The random-walk benchmark (§2)**. The rate argument is right: $\hat{\beta} = O_p(1)$, $\text{se} = O_p(T^{-1/2})$, $|t| = O_p(T^{1/2})$, rejection probability $\rightarrow 1$.
- **All λ values and asymptotic benchmarks in Table 1**. I recomputed each independently: 1.667/0.129, 4.556/0.358, 9.526/0.525, 19.513/0.657, 2.636/0.227, and $0.220/2.87 \times 10^{-5}$. All match.
- **Every rejection frequency quoted in §5.1–5.5** against the corresponding CSV: 12.6, 34.8, 51.4, 66.0, 21.6% and zero rejections for (0.8, -0.8); HAC 10.7% for (0.95, 0.95); local-to-unity

45.9% → 94.1% conventional and 31.9% → 71.6% HAC; Granger 7.2%, 30–34%, 7.2%. All correct.

- **The MCSE calculation** $\sqrt{0.25/2000} \approx 0.0112$. Correct.
- **The AR(1) initialisation.** `stats::filter(..., init = rnorm(1, sd = 1/\sqrt{1-\phi^2}))` yields $x_1 \sim N\{0, 1/(1 - \phi^2)\}$ exactly. The stationarity claim in §4 holds.
- **The reshape/match column construction** in both `tbl-conventional` and `tbl-hac`. Fragile (positional `names()` assignment), but correct as written — I checked the rendered values against the CSVs cell by cell.

One arithmetic correction. §5.3 gives the asymptotic value implied by $\lambda = e^{0.5}$ as 12.8%. It is $2\Phi(-1.96/\sqrt{e^{0.5}}) = 0.1269$, i.e. **12.7%**. The simulated 12.35% is closer to theory than the paper claims.

Recommendation

Major revision.

M1, M2, M5, P1 and P2 are errors and must be corrected before the paper can be considered further; none is difficult. M4 is a matter of scholarly positioning and is the difference between a paper that reads as unaware of its own literature and one that engages with it.

Beyond that, the honest question is what the paper contributes. The λ characterisation and the AR(1) product formula are established results, and as it stands the paper is a careful, well-executed re-derivation with a clean simulation. That has real pedagogical value — I would happily assign it — but it is not yet a contribution. The three additions that would change that are, in order of value: **local power** (M6), a **non-AR(1) DGP demonstrating correct size under visible serial correlation** (M7), and **bandwidth/kernel sensitivity with fixed- b critical values** (M8, M4). Each is a modest extension of machinery the author already has running. Together they would turn a good exposition into a paper with something of its own to say.

The writing is precise and the author is commendably careful to state what the results do *not* show — the caveats in §2, §3.6 and the Discussion are exactly right in spirit. I would encourage the author to extend that same scepticism to the two claims flagged in M1 and M2, where the prose currently outruns the evidence the paper itself presents.